

Fake Image Detection in Fake News Using Convolutional Neural Network (CNN)

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Abstract—'Fake news' refers to false, inaccurate, or misleading information that spreads as real news. Fake news primarily aims to affect societies and individuals by spreading false or misleading information. Fake news comes in text, images, videos, and audio formats. Detecting fake news is a research challenge that requires a promising solution to assure the accuracy of the information. Recently, promising results have been generated using artificial intelligence to detect fake news in multimedia. Computer vision, natural language processing techniques, audio processing, and machine learning algorithms are all techniques used in artificial intelligence to help accurately differentiate accurate news from fake news. However, The paper proposed a Convolutional Neural Networks (CNNs) model that can detect fake images in the context of fake news. The paper also utilized various preprocessing techniques, including error level analysis (ELA), gradient, and noise analysis. These techniques are compared to discern their effectiveness in enhancing the model's performance. The result indicates that the proposed CNN model achieved an overall accuracy of 91.38% on the CASIA V2.0 dataset with error level analysis (ELA) as the preprocessing technique. The results indicate that ELA is an effective preprocessing technique for extracting features that can be used to identify fake images.

Index Terms—Fake image detection, Fake images, Convolutional neural network, Deep learning

I. INTRODUCTION

Nowadays, the manipulation of digital images has become effortlessly accessible; sophisticated picture editing software and high-resolution imaging equipment have made it easy to manipulate digital photos. This ease of manipulation raises concerns about the authenticity and reliability of image content, emphasizing the need for robust solutions to address the challenges posed by manipulated images in various domains. With the widespread proliferation of digital images, the accessibility of tools for manipulating and distorting them has also been easy. Consequently, this manifests in various multimedia formats, text, images, videos, and audio, amplifying its impact and reach.

The growth of image processing software has led to the growth of fake images; in most cases, the human eye cannot recognize these fake images, necessitating the development of fake image detection algorithms. These algorithms should be able to accurately determine whether an image is real without relying on basic information about its content. The demand for fake image detection extends to various fields, including legal-judicial investigations, publishing, news agencies, criminal investigations, insurance, medical imaging, and scientific research. This underscores the critical role of fake image detection algorithms in addressing the challenges of manipulated visual content across diverse domains [1].

Detecting fake images within the context of fake news presents a research challenge that intersects the domains of image processing, computer vision, and artificial intelligence. The manipulation of images to support false claims or misleading stories has become a ubiquitous challenge, making detecting fake images a critical concern. However, artificial intelligence has been proven to help detect fake images, as mentioned in multiple studies, with promising results.

This paper presents a model Convolutional Neural Networks (CNNs) to detect fake images within the context of fake news. Pre-processing techniques are applied to the input images as a preparatory step before their integration into the CNN model. Several techniques, including error level analysis (ELA), gradient analysis, and noise analysis, are employed for this goal. The application of these techniques is compared to discern their respective efficacy, aiming to identify the most effective approach for enhancing the model's performance. This comparative analysis contributes to optimizing the overall image detection process within fake news detection.

The paper is structured as follows: Section II defines the problem statement and research questions, Section III presents the related work, including the single modality approaches and multi-model approaches, and Section IV defines methodology, including dataset, Pre-processing techniques, Convolutional

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Neural Network (CNN), and evaluation matrices. Section V demonstrates the results and discusses them. Finally, it concludes and presents future work in Section VI.

II. PROBLEM STATEMENT AND RESEARCH QUESTIONS

Recently, fake news has seen vast growth through multimedia containing images and videos. This issue has become a major global issue because fake news can seriously affect individuals and society. Therefore, developing effective techniques to detect fake news in images is essential. Artificial intelligence has emerged as a promising solution for detecting fake news in multimedia. The paper aims to address the issue of fake news in images, concentrating on images as primary content sources. The paper aims to utilize image processing techniques to distinguish signs of manipulation or fake images and to improve and enhance the accuracy of fake news detection. These techniques involve extracting different image features to determine the reality of the image content. The paper seeks to answer these questions:

- 1) What are the most effective image processing techniques for identifying fake images?
- 2) What are the key challenges in using Error Level Analysis (ELA), Noise Analysis, Gradient Analysis, and Convolutional Neural Networks (CNNs) to effectively detect fake images in fake news?
- 3) What is the accuracy of the proposed CNN model for fake image detection in fake news?

III. LITERATURE REVIEW

Recently, fake news research has received much attention due to the rise of social media and the spread of misinformation online. Therefore, this indicates the need to develop effective ways to detect fake news. Classifying between real and fake news is increasingly difficult, especially when images are involved. Existing approaches for detecting fake images in fake news using deep learning can be split into single-modality and multi-model approaches [2]. This section shows the recent research on fake image detection in fake news using deep learning.

A. Single modality approaches

Single modality approaches mean only a single data type, such as text, image, audio, or video, is utilized to classify the news as fake. Ghannad and Passi [3] suggests an approach using U-NET with a grasshopper optimization algorithm (GOA) for detecting fake images. The approach utilizes the U-NET network for image segmentation and then applies the GOA to enhance the accuracy of the results. The approach architecture is developed with an encoder and a decoder, along with convolutional and max pooling/upsampling layers to help detect the fake area in images. After training, the GOA is applied to enhance the U-NET network. The proposed approach U-Net with GOA achieved a high accuracy of 99.1% with a train/valid ratio of 8:2. However, their implementation is computationally expensive and lacks usability in real-time systems.

Furthermore, this paper Kasban and Nassar [4] introduces a highly accurate method for detecting digital image forgery, involving several key steps: firstly, it transforms the RGB image into the YCbCr color space and further employs the Hilbert-Huang transform to extract features from the chrominance-red component (Cr). Next, it conducts image classification using three distinct classifiers, namely Support Vector Machines (SVM), Artificial Neural Networks (ANN), and K-Nearest Neighbors (KNN). These classifiers are employed and compared to categorize images as real or fake. The findings indicate that the SVM classifier outperforms the ANN and KNN classifiers, achieving the highest accuracy. Finally, the results are validated through the Structural-Similarity (SSIM) to assess the accuracy of forgery detection. Nonetheless, because these methods require substantial storage for the training samples, it can considerably impede the process. Furthermore, it only supports binary classification.

Tyagi and Yadav [5] proposed MiniNet is a simple and efficient architecture for detecting fake images. The MiniNet architecture is mainly helpful in detecting two fake image types: Copy-move and Splicing. The proposed architecture addresses two limitations of existing image fake detection models: it can identify forgeries regardless of image dimensions and is resistant to tampering attacks. MiniNet, a CNN with eight layers, also includes four convolutional layers and employs (2×2) max-pooling layers to reduce feature map sizes. MiniNet is trained and assessed on publicly available datasets, 140k Real and Fake Faces (RFF) and CASIA, demonstrating an accuracy exceeding 95% and 93%, respectively.

Moreover, other CNN with error-level analysis were proposed [6, 7]. Kubal et al. [6] introduce an EACN system that combines error-level analysis and CNNs to detect fake images. The proposed technique is designed to distinguish if the image is fake by generating a new feature image derived from subtracting between the original and recompressed images. The ELA approach is more powerful at identifying image manipulation. The CNN model used in EACN has fewer convolutional layers (only three layers and, after each, a max-pool layer) and parameters compared to existing approaches by other authors, allowing for convergence in less than ten epochs. The authors complemented their approach by applying error level analysis to quantify the error rate resulting from reduced image quality. The feature image, generated by subtracting the recompressed image from the original, was the basis for training the CNN model to determine whether the image was fake. They evaluated the system on CASIA 2.0 datasets and achieved an accuracy of up to 92.10% in accurately detecting fake images.

Furthermore, Shah et al. [7] proposed an approach to a deep learning-based method for detecting picture splicing in photographs. The proposed approach involves two key steps: ELA Preprocessing: The input image undergoes preprocessing using an "Error Level Analysis (ELA)," which extracts noise remnants by suppressing image content. Furthermore, feature Extraction and Classification: The well-known MobileNet network is employed as a feature extractor, and the obtained

features are examined for authenticity using the voting classifier. Moreover, experimental results using the CASIA dataset demonstrate that the proposed method outperforms existing approaches. It achieves an average classification accuracy of 94.24%, showcasing its effectiveness in detecting picture splicing in images.

Moreover, Xue et al. [8] presents a Multi-Vision Fusion Neural Network (MVFNN) comprising four primary modules: the ensemble module, the visual modal module, the visual feature fusion module, and the physical feature module. The visual modal module extracts image characteristics from several domains, including pixel, frequency, and tamper detection. It uses multi-vision fusion with the visual feature fusion module to identify bogus news photos. The ensemble module combines physical and visual features to identify fraudulent news photos. The model's validity is verified through comprehensive experiments on two datasets. The first dataset (D1) is sourced from the Zhiyuan Fake News Recognition Competition and includes 20,459 real news pictures and 13,636 fake news images, all originating from news on Weibo. The second dataset (D2) comprises 3,725 real news pictures and 2,804 fake news pictures sourced from reputable and authoritative news websites. Furthermore, it employs these established methods to evaluate the effectiveness of the proposed fake news picture detection method: Forensics features (FF) + LR, Pre-trained VGG19, and ConvAE. Experimental results indicate that this model outperforms existing methods by a minimum of 4.29% in benchmark datasets, demonstrating superior detection performance.

B. Multi-modal approaches

In recent years, researchers have developed multi-modal approaches for fake news detection, which have been shown to outperform single-modal methods. A multi-modal approach uses multiple data types, such as text with images or videos. Singh and Sharma [2] presents a multimodal method to detect fake social media images. This method utilizes images and text features to predict the credibility of an image. The model that has been suggested comprises of two primary components, namely an image feature extractor and a text feature extractor. For the image feature extractor, the EfficientNetB0 model has been utilized, which has been demonstrated to have higher accuracy as compared to other CNN models for image classification. The text feature extractor is based on the sentence transformer RoBERTa model, which is bidirectional encoder-based and used for text classification. The text and image feature sets obtained from both modalities are fused together, and then two dense layers are added to learn these feature sets. Once the input has been processed, it is fed into the softmax classifier layer, which predicts the credibility of an image. The proposed model is evaluated on CASIA 2.0, MediaEval, and Chinese Weibo datasets. The suggested model achieves an accuracy of 87.13% on the CASIA 2.0 dataset, 85.3% on the MediaEval dataset, and 81.2% on the Weibo dataset. The results demonstrate that the proposed model effectively detects fake images, outperforming other multi-modal frameworks.

Also, Raj and Meel [9] suggests a new fake news detection framework. ConvNet framework incorporates textual and visual input streams for more accurate classification. ConvNet frameworks consist of Text-CNN and Image-CNN, one for text fake news and one for visual fake news classification. Text-CNN embeds words into vectors, applies convolutional layers with filters to generate features, uses max-pooling to select the most salient features, and passes them through fully connected layers for classification. On the other hand, Image-CNN utilized various CNN models such as AlexNet, ResNet50, DenseNet, MobileNet, Xception, InceptionV3, VGG-16, and VGG-19 for image classification. The Image-CNN module analyzes the latent features in the images to classify them as real or fake. The proposed model is evaluated on Ti-CNN, Emergent, and MICC-F220 datasets. The highest accuracy was achieved with Text-CNN with VGG16 with 98.93% and 94.05% accuracy scores on different datasets. After analyzing different CNN models on datasets, they have proven that VGG16 consistently performs well, followed by Xception and MobileNet.

Furthermore, the SpotFake system was proposed by Singhal et al. [10], a multimodal framework that relies solely on an article's text and image features, which is divided into three sub-modules. The initial sub-module serves as a textual representation, and features are extracted using language models BERT—the second sub-module acts as a visual feature extractor. Image features are obtained by applying VGG19 pre-trained on the ImageNet dataset. Finally, the last sub-module functions as a multimodal fusion module, combining the representations brought from various modalities to construct a new feature vector. The Twitter and Weibo datasets experiments demonstrate the model accomplishes highest result than the current state-of-the-art methods, with improvements of 3.27% and 6.83%, respectively. Moreover, model accuracy SpotFake Twitter: 77.77%, Weibo: 89.23%. Limitations for VGG-19 (CNN model): It takes an increased training duration, requires a larger dataset for improved generalization, and entails more hyperparameters during the training process.

Nevertheless, there is also some proposed work that combines CNN and ELA. Such as, this paper Masciari et al. [11] presents a framework for image analysis in fake news detection, highlighting a thorough method based on the NLP module that uses Google BERT and TF-IDF to detect bogus text. It addresses fake images by combining CNN and ELA at the same time. The activation function is called ReLU, and the optimizers are Momentum, RMSProp, and Adam. Softmax and sigmoid are included in the final layer for binary classification. The ideal batch size for the CNNs is 32, and they are trained for ten epochs. Furthermore, the framework uses a diversity of data sources. The dataset used is called FakeNewsNet, and it consists of data that was taken from GossipCop and Politifact. Moreover, the framework might be adapted to include the LIAR and PHEME datasets. Additionally, the proposed approach performed consistently when applied to different datasets, achieving an accuracy rate of 76.5% on the PHEME dataset.

TABLE I: Literature summary

	Paper	Technique	Dataset	Accuracy
Single Modality Approaches	Ghannad and Passi [3]	U-NET network and GOA	CASIA	99%
	Tyagi and Yadav [5]	CNN	CASIA and 140K Real and Fake Faces	93% and 95% respectively
	Kubal et al. [6]	ELA and CNN	CASIA	92%
	Kasban and Nassar [4]	Hilbert-Huang transform (HHT)	CASIA V1.0 and V2.0	94.09%
	Shah et al. [7]	ELA and MobileNet network	CASIA	94.24%
	Xue et al. [8]	Multi-Vision Fusion Neural Network (MVFNN)	Weibo	~
Multi-modal Approaches	Singh and Sharma [2]	EfficientNetB0 and RoBERTa	CASIA V2.0, MediaEval, and Weibo	87.13%, 85.3%, and 81.2% respectively
	Raj and Meel [9]	CNN	Ti-CNN, Emergent, and MICC-F220	The combination of Text-CNN with VGG16 produces the best accuracy scores of 98.93% on the TI-CNN dataset and 94.05% on the Emergent dataset
	Singhal et al. [10]	VGG19 and BERT	MediaEval, and Weibo	77.77% 89.23%
	Masciari et al. [11]	ELA and CNN	FakeNewsNet	76.5%

In conclusion, this section reviews the latest studies on single-modality and multi-modal approaches in fake image detection in fake news. The table I summarizes each study's technique, dataset, and accuracy. While multi-modal approaches show promise and bright solutions in fake image detection in fake news, they come with limitations. Firstly, they tend to be computationally intensive and require substantial computational resources, making them less accessible for researchers with limited computing capabilities. Secondly, it often requires large and mixed datasets to train the model. Multi-modal approaches can be complicated, more complex, and computationally expensive to train and evaluate. Furthermore, some existing solutions that depend on CNN have shown high performance. However, the existing research solutions have some drawbacks, such as using many layers, training slowly, and having complex implementation.

In addressing these limitations, this paper suggests a single-modality approach for fake image detection in fake news, aiming to provide a simpler solution that is more efficient and less computationally expensive than existing methods. This paper introduces a model that compares three different preprocessing techniques with the CNN model to detect fake images within fake news. CNN then learns to extract features from the three different preprocessing techniques for images and then can classify the images as fake or not. Furthermore, the proposed model leverages the capabilities of neural networks and will optimize the model for efficient training and achieving high accuracy.

IV. METHODOLOGY

The ability to distinguish between real and fake images is crucial in today's world, especially in the context of fake news. Therefore, the main objective of this paper is to create a robust and accurate model that detects fake images effectively. Fake images are images that have been manipulated or altered in some way to mislead the individual and society. Fake news is misinformation intentionally or unintentionally spread through various media platforms. To achieve this goal, the proposed

methodology for fake image detection in fake news employs pre-processing techniques and convolutional neural networks (CNN) to classify images as fake or real. Figure 1 illustrates the proposed methodology.

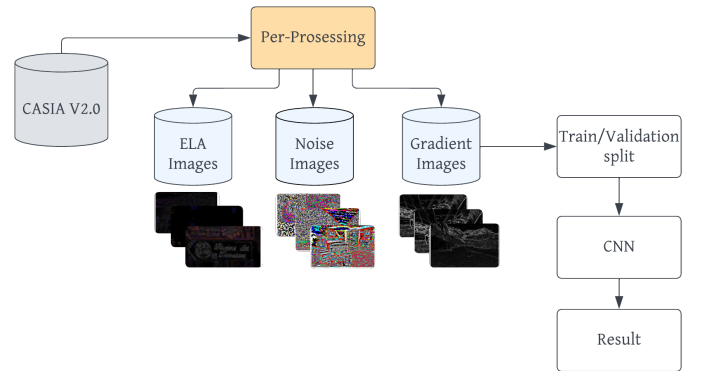


Fig. 1: The Proposed Methodology

A. Dataset

The dataset utilized for training the model is the CASIA V2.0 [12], a benchmark forgery classification dataset. The CASIA V2.0 dataset comprises 12,617 images, 7,492 real and 5,125 fake. Figure 2 shows samples of the dataset.

B. Pre-processing Techniques

Three preprocessing techniques were employed to generate new datasets for feeding into the proposed model. The pre-processing techniques can help extract features that can help identify the fake images. Once the data was preprocessed, it was divided into two sets based on an 80:20 ratio, training set, and validation set, with random images in both. The main goal of the training set is to train the model, and the validation set is to assess and evaluate the model's performance. Three preprocessing techniques are:

- **Error Level Analysis (ELA):** Given the lossy nature of the JPG algorithm, which introduces new compression

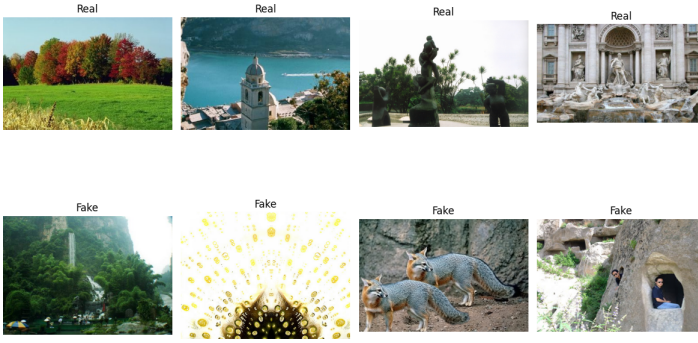


Fig. 2: Samples of CASIA V2.0 Dataset

errors with each save, the ELA algorithm is developed to identify varying error rates in the image. ELA works by re-saving the image at a specified quality level and then computing the difference between the original and the re-saved image. Regions with higher differences in error levels are potential indicators of possible fake images [6, 7].

- **Noise Analysis:** This technique relies on identifying alterations in the inherent natural noise present in each image from different factors such as the lens, lighting conditions, or other digitalization techniques. The approach is designed to detect the changes in the natural noise pattern, which may arise when manipulating the image. Noise analysis works by applying a high-pass filter to the image and then computing the standard deviation of the filtered image. The regions with lower or higher standard deviation than the average indicate possible fake images.
- **Gradient Analysis:** This technique involves identifying variations in brightness levels throughout the image and identifying any anomalies. Gradient analysis works by computing the gradient magnitude and direction of the image and then applying a threshold to detect the edges. The regions with discontinuous or inconsistent edges indicate possible fake images.

C. Convolutional Neural Network (CNN)

The CNN is an artificial neural network containing numerous neuron layers. These neurons can process the input data hierarchically. The CNN model can learn features from the data by applying convolutional filters, pooling operations, and activation functions. The CNN model can also perform classification tasks using fully connected layers and softmax functions. CNNs are capable of extracting complex patterns and features from images, which makes them ideal for detecting fake images [5, 6, 9, 11]. When the CNN is trained on a large dataset of real and fake images, it distinguishes between the two classes and makes accurate predictions. The proposed CNN model for fake image detection in fake news incorporates two convolutional layers, one max-pooling layer, one dropout layer, one flatten layer, and two dense layers. The convolutional layers are responsible for extracting the features

of the input images, such as edges, shapes, textures, and colors. The dense layer is used to reduce the dimensionality of the features and learn the non-linear relationships between them. The output layer uses the binary cross-entropy loss function. Moreover, the Adam optimizer was used to classify the images as real or fake. The proposed CNN model also incorporates dropout layers to avoid overfitting and enhance the generalization performance.

Figure 3 summarizes the proposed CNN architecture. The following is a detailed description:

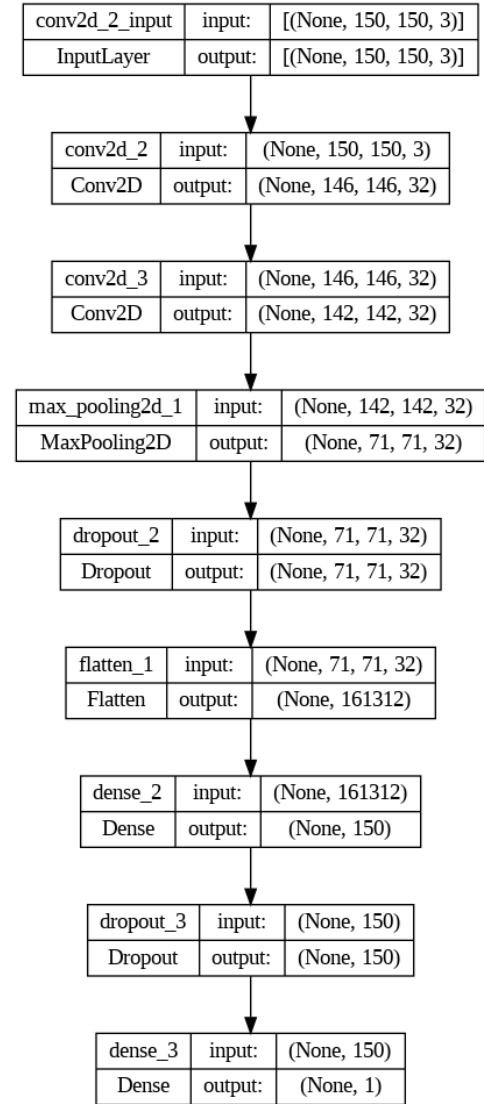


Fig. 3: Architecture of the proposed CNN model

- 1) Convolutional Layer (Conv2D): The process's initial step is to extract local and core features from the input images. The input images are subjected to a first convolutional layer comprising 32 filters with a kernel size of 3x3. The resulting feature maps are then passed through a rectified linear unit (ReLU) activation function.
- 2) Convolutional Layer (Conv2D_1): This layer extracts

more complex features from the feature maps produced by the first convolutional layer. The second convolutional layer also has 32 filters with a kernel size of 3x3 and applies them to the input feature maps. The resulting feature maps are then passed through another ReLU activation function.

- 3) Max Pooling Layer (MaxPooling2D): This layer reduces the dimensionality of the feature maps by downsampling them by a factor of 2. This helps to reduce the number of parameters in the model and can also help to improve the model's generalization performance.
- 4) Dropout Layer (Dropout): This layer helps to prevent overfitting and improve the model's generalization performance by randomly eliminates a particular percentage of neurons in the preceding layer.
- 5) Flatten Layer: This layer flattens the output feature maps from a 2-dimensional into a one-dimensional vector.
- 6) Dense Layer (Dense): This fully connected layer takes the flattened feature vector as input and produces 150 output neurons. The ReLU activation function was used.
- 7) Dropout Layer (Dropout_1): This layer randomly eliminates a particular percentage of the neurons in the preceding dense layer.
- 8) Dense Layer (Dense_1): The model's last layer has a single output neuron and applies the sigmoid activation function to it.

D. Evaluation Metrics

Four evaluation metrics were considered to measure the proposed model's performance:

- **Accuracy:** This metric simply indicates the proportion of predicted image classifications that were correct out of the total number of images.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

- **Precision:** This metric measures the proportion of all positive predictions (classified as real) that were actually real.

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

- **F1 score:** This metric is calculated as the harmonic mean of precision and recall.

$$F1 = \frac{2 * (Precision * Recall)}{(Precision + Recall)} \quad (3)$$

- **Recall:** This metric measures the proportion of actual real images that were correctly classified as real.

$$Recall = \frac{TP}{TP + FN} \quad (4)$$

V. RESULTS & DISCUSSION

This paper proposed a CNN model for fake image detection in fake news on the CASIA 2.0 dataset. The proposed model employs three different pre-processing techniques on the proposed CNN architecture to extract features from the input

TABLE II: Evaluation metrics of the proposed model on Validation Set

Pre-processing Techniques	Loss	Accuracy	Precision	Recall	F1-score
ELA	0.2894	0.9138	0.9468	0.8834	0.9140
Noise	0.5466	0.7607	0.7850	0.7481	0.7661
Gradient	0.9764	0.6824	0.7625	0.5616	0.6468

TABLE III: Evaluation metrics of the proposed model on Train Set

Pre-processing Techniques	Loss	Accuracy	Precision	Recall	F1-score
ELA	0.0684	0.9760	0.9756	0.9756	0.9756
Noise	0.3005	0.8582	0.8981	0.8296	0.8625
Gradient	0.2070	0.9375	0.9265	0.9450	0.9356

images and classify them as real or fake. The pre-processing techniques used are error level, noise, and gradient analysis. These techniques were chosen based on their ability to extract features that distinguish between real and fake images. The proposed CNN model for fake image detection in fake news incorporates two convolutional layers, one max-pooling layer, one dropout layer, one flatten layer, and two dense layers. The convolutional layers extract meaningful features from the input images. The dense layer is used to reduce the dimensionality of the features and learn the non-linear relationships between them. The output layer is used to classify the images as real or fake. The proposed model also utilizes dropout layers to prevent overfitting and enhance the generalization. The evaluation metrics to assess the proposed model were accuracy, recall, precision, and F1 score.

The CNN model was trained on three datasets (ELA, Noise, Gradient) created from the preprocessing step, each divided into 80:20 training and validation sets. This allowed the model to better generalize to unseen data. The proposed CNN model was trained for 30 epochs, with streamed images set to 150 x 150 and a batch size 8. These optimized settings were key in achieving better results during the training process.

Table II and III summarize the result of validation and training sets, respectively, for each preprocessing technique. The results indicate that the ELA demonstrated the highest accuracy, with a validation score of 91.38% and a training score of 98%. The noise came in second, with a validation score of 76.8% and a training score of 86%. Meanwhile, the gradient had the lowest accuracy, with a validation score of 68.24%, but the training score of 94% was higher than noise accuracy in training.

Precision measures the proportion of true positive predictions among all positive predictions. The ELA showcased a precision rate of 97.56% and 94.7% in both training and validation sets, suggesting correctness when identifying images as fake. Nevertheless, noise and gradient demonstrated lower precision values of 79% and 76.25%, respectively, in validation. However, in the training set, the gradient achieved a higher precision of 93%, while noise was slightly lower at 90%.

The recall assesses the model's ability to catch all actual

positive samples. In training, the ELA demonstrated the highest recall, achieving an impressive value of 97.56%, while in validation, it reached 88.34%. Conversely, the Gradient technique exhibited the lowest recall, reporting at 56.16%. Additionally, noise showed the lowest recall in the training set, with a value of 82.96%.

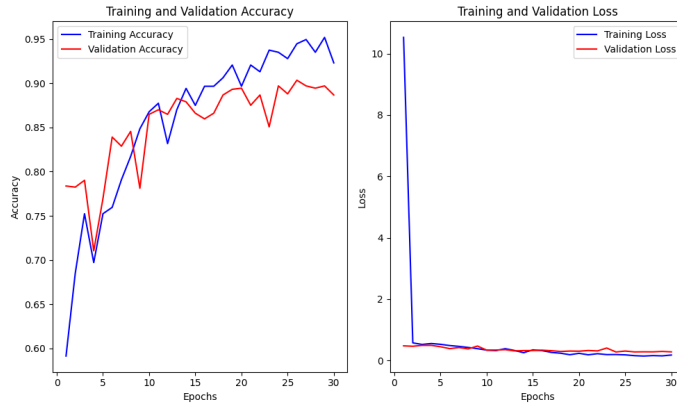


Fig. 4: Training and Validation Accuracy and Loss for ELA



Fig. 5: Training and Validation Accuracy and Loss for Noise

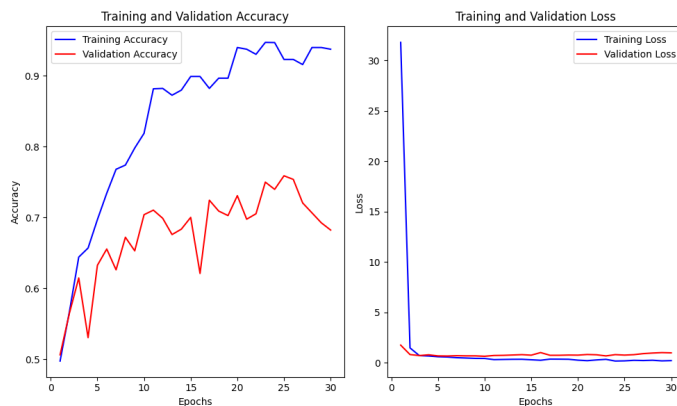


Fig. 6: Training and Validation Accuracy and Loss for Gradient

The F1 score it considers both precision and recall, providing a single metric that balances both performance metrics. The ELA outperformed other techniques in the validation set, with the highest F1 score of 91.40%. In the training set, the ELA value is 97.56%. Conversely, the gradient technique exhibited the lowest F1 score in the validation set, a value of 64.68%. Notably, noise demonstrated the lowest F1 score in the training set at 86.25%.

Figures 4, 5, and 6 demonstrate a graph of training and validation loss and accuracy for the proposed CNN model trained on three different datasets: ELA images, noise images, and gradient images, respectively. The training accuracy graph indicates that the accuracy is increasing over time, while the training loss graph illustrates that the training loss decreases over time, which indicates that the model is learning and it improving the ability to categorize the correct output.

Furthermore, the validation accuracy graph shows that accuracy increases over time but at a slower rate than training accuracy. This is expected, as the validation set is held out from the training process and is used to assess the proposed CNN model's performance on unseen data. The fact that the validation accuracy is also increasing indicates that the model generalizes well to unseen data. The validation loss graph shows that the loss also decreases over time but at a similar rate to the training loss. This is also expected, as the validation loss is calculated using the same loss function as the training loss.

The graph shows that the CNN model trains well and generalizes well to unseen data. Furthermore, the training and validation accuracy curves are close together, indicating that the model does not fit the training data. The validation accuracy curve is increasing steadily, indicating that the model generalizes well to unseen data. The validation loss curve is decreasing steadily, indicating that the model can learn from the training data and improve its capability to indicate the correct class on the validation set.

Nevertheless, some key challenges exist in using error level analysis (ELA), noise analysis, gradient analysis, and CNN to detect fake images in fake news [13, 14]. ELA, noise analysis, and gradient Analysis are based on pixel-level or compression-level features, which may not be robust or reliable enough to capture the semantic or contextual information of the images [14]. As a result, shows, the ELA sometimes failed to detect tampering in low-quality or high-compression images, noise analysis was affected by different noise sources or levels, and gradient analysis was sensitive to threshold or filter parameters; this is explained it was the noise and gradient has the lowest accuracy. Moreover, these techniques can't handle the diversity and complexity of the fake images.

After conducting a thorough analysis and evaluation, it is evident that ELA stands out as the best-performing pre-processing technique with the proposed CNN. This conclusion is based on several metrics, including accuracy, precision, recall, and F1 score, where ELA performed significantly better than other pre-processing techniques. These results demonstrate the robustness and reliability of ELA in detecting fake

images on both the validation and training sets.

VI. CONCLUSION & FUTURE WORK

Fake news is a big problem that can mislead, manipulate, and harm the individual and society. Fake news can be spread through various media, such as text, audio, video, and images. Among them, images are one of the most influential and effective forms of communication, as they can capture the attention and emotions of the individual. This paper proposed a CNN model for detecting fake images in fake news. The proposed CNN model consists of three convolutional layers (including the max pooling layer), one dropout layer, one flatten layer, and two dense layers. The proposed model is simpler and less computationally expensive due to the low number of layers. The findings showed that the ELA outperformed other techniques regarding accuracy, precision, recall, and F1 scores—an accuracy of 91.38% using ELA on the CASIA 2.0 dataset. Furthermore, this paper contributes to the fake image detection area by evaluating and comparing the performance of various pre-processing techniques. The results indicate the effectiveness of ELA in accurately identifying fake images. Future work will explore using other pre-processing techniques and CNN architectures for fake image detection. Another suggestion for future work is to evaluate and assess the proposed model on larger and more various datasets.

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